

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. The first supplies neither of the others, and four results establish that without any cortical hypothesis. Synchrony constrains content only through encoders shared on overlaps, and then by the resultant of the patch the bound is read on rather than of the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable. A contraction with a unique fixed point can discard its input, so accuracy across a family of states requires distinguishable codes. And where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely, and installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it

to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices; energy is not the scarce side of that condition for a macroscopic substrate, which leaves it a statement about κ , derived here for neither cortex nor a processor. Lean 4 checks these results and their composition, with the eight connecting hypotheses as explicit arguments and no axioms beyond its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, measured here as $\Delta K \propto v^{0.444}$, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their

domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 9.4 returns to the comparison.

Consider reaching for a cup. A visual description of its location and a description of the hand’s relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and fields in real normed vector spaces, whose identification with perceptual content is a further modelling task.

The mathematical contribution is to specify connections between these requirements and the assumptions each connection needs. A declared actuator’s stored energy bounds its mean coupling. Phase encoders shared on

overlaps convert phase order into a bound on content disagreement. Compatible spatial measures have a unique global extension; approximate agreement controls a specified global selection. A contracting reconstruction makes that extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition bounds mean coupling by the stored energy of a declared actuator, which makes a resource condition necessary for coherence on the stationary branch. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps

58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis
80 of Section 8 still has to accommodate them. Sections 3–5 then develop the

81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support.

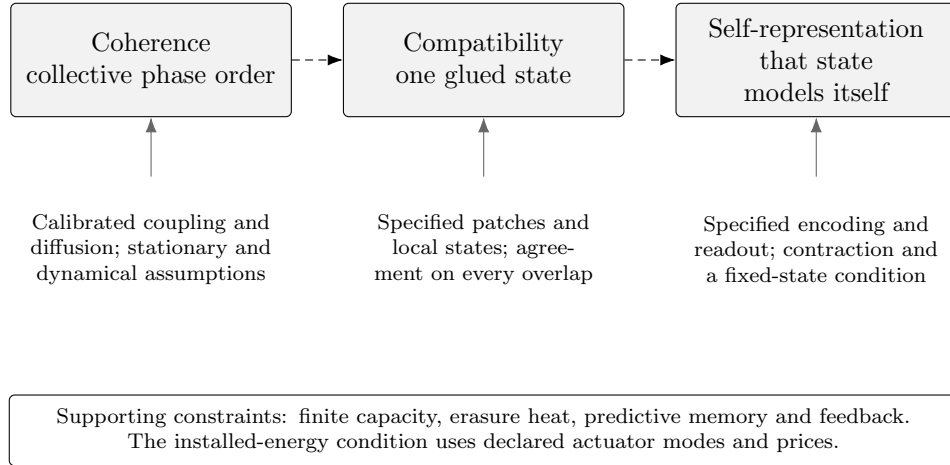


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them.

117 **A unique fixed point is not a self-model.** A constant readout con-
 118 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 119 is consequently the wrong criterion, and Section 5.2 gives the right shape
 120 of one: reconstructing M states pairwise more than 2ε apart within error ε
 121 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 122 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 123 readout whatever.

124 **Where the encoding is a restriction, the region must resolve the**
 125 **family.** Read the code as the state's own restriction to a region and
 126 the criterion moves off the readout entirely. A boundary whose region
 127 reads two relevant states identically cannot reconstruct them more than 2ε
 128 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
 129 boundary that does reconstruct a separated family restricts injectively on
 130 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
 131 region sits is fixed before an encoding is chosen, so no readout supplies
 132 resolution the region lacks.

133 **Agreement after a change takes communication.** In a synchronous
134 network whose updates read only the site’s own state and its declared neigh-
135 bours’ messages, two configurations agreeing on a site’s causal past through
136 T rounds leave that site identical at round T . Two interventions outside
137 that past that require reports more than 2ε apart therefore cannot both be
138 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
139 ity and timely compatibility are therefore different requirements, and cutting
140 the path removes the second while leaving the first intact.

141 None of these depends on the coupling being electromagnetic, on the
142 substrate being cortex, or on the eight hypotheses of Table 1 holding. Sec-
143 tion 9.3 applies the last three to a candidate that is neither biological nor
144 field-coupled.

145 **3 Coherence: physical coupling and its re-** 146 **source condition**

147 **3.1 Dynamics and the stationary threshold**

148 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

149 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 150 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
 151 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
 152 parameters. The stationary calculation below uses identical frequencies and
 153 this scalar coupling. Applying it to a spatial kernel requires an aggregation
 154 rule and a check of the resulting approximation.

155 For identical frequencies in their rotating frame, the positive normalized
 156 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

157 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 158 mal derivation uses the absence of stationary current for a periodic gradient
 159 drift to obtain this density, then verifies normalization and self-consistency.
 160 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 161 threshold, exactly one positive solution coexists with zero and increases
 162 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

163 Figure 2 shows this stationary branch.

164 Linearization about the uniform density gives rate $K/2 - D$ for the first
 165 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 166 exactly above threshold. About the coherent density, the quadratic form

167 is strictly dissipative transverse to collective rotation, whose rate is zero.
 168 These linear estimates concern the limiting equation. Relating them to finite
 169 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 170 tion; exchange symmetry alone does not establish the needed independence
 171 (Sznitman, 1991).

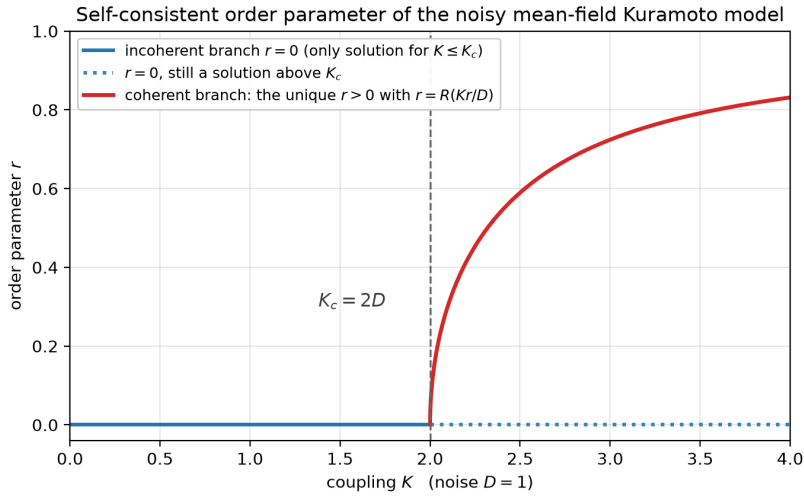


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

172 3.2 Installed energy bounds mean coupling

173 A spatial model replaces the finite sum by integration against a kernel. One
 174 declared implementation carries response modes with continuous profiles ϕ_i ,

175 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

176 The same configuration sets the reciprocal kernel and stored energy. An
 177 executed transition changes the kernel through its changed occupancies and
 178 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 179 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 180 pair reconstruction on refining spatial partitions converges to its product-
 181 measure integral under the specified compactness and sampling assumptions;
 182 the refinement index is independent of the number of control updates.

183 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 184 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 185 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

186 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 187 coherent stationary branch and any growing Fourier mode about the uniform
 188 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 189 is necessary: energy above it need not produce sufficient coupling because
 190 the bound can be loose.

191 The quantity constrained is installed energy rather than dissipated heat,
 192 and how much the condition excludes depends entirely on κ , which the mode
 193 profiles, prices and substrate normalization supply. Any macroscopic sub-
 194 strate carries installed energy far above the thermal floor, so the inequality
 195 binds through that conversion and not through the energy, and no result here
 196 derives κ for cortex or for any other candidate; what the condition contributes
 197 is a resource requirement in units a measurement can reach. Applying the
 198 scalar threshold to heterogeneous or signed spatial dynamics requires a jus-
 199 tified reduction, and identifying the installed kernel with a cortical field is a
 200 separate physical commitment.

201 **4 Compatibility: from coordinated activity** 202 **to a global description**

203 **4.1 Why common timing is insufficient**

204 For overlapping patches U and V , compatibility means that restricting either
 205 local description to $U \cap V$ gives the same object. Descriptions can agree on
 206 shared quantities while contributing different information elsewhere. Com-
 207 patible sections of the specified sheaf have a unique global extension. On
 208 a finite cover, local finite spatial measures agreeing as measures on every
 209 overlap also have a unique finite-measure extension whose sheaf image is the
 210 glued section. No mass-one normalization is imposed.

211 In the cup example, visual and reaching descriptions must express a
 212 shared quantity, such as cup location, in common coordinates. Restriction
 213 retains that quantity while dropping details particular to each description.
 214 Agreement permits surface appearance and hand configuration to contribute
 215 to one situation. The mathematical counterexample assigns the same phase
 216 to every patch while two local mass profiles disagree on a shared site. Those
 217 profiles have no common extension. Because they are chosen independently
 218 of phase, synchrony places no constraint on their disagreement.

219 This distinction limits the biological interpretation. Receptive territories
 220 and content domains must be related to spatial patches, and the descriptions
 221 must support the required extension. Matching overlap marginals of de-
 222 coded probability laws need not determine a unique global joint distribution
 223 (Abramsky & Brandenburger, 2011).

224 **4.2 When coherence does constrain content**

225 A declared relation between phase and content gives a positive result. Let
 226 contents take values in a real normed vector space E . Patch i reads an
 227 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 228 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 229 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 230 sites; their equality on overlaps is a supplied relation between descriptions of
 231 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 232 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

233 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

234 Here discrepancy is the norm of the difference, and r and N are read on
 235 whichever population the bound is applied to. That choice is what decides
 236 whether the estimate says anything. A patch carries an order parameter of
 237 its own, and reading the bound there carries that patch's resultant and that
 238 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 239 `coherence`), so a large population observed through small patches is con-
 240 strained by the small count. Read instead on the whole array, the factor N
 241 makes the estimate weak, because it draws a pointwise guarantee from an
 242 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 243 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 244 tion permits disagreement even at perfect locking. A common nonexpansive
 245 linear projection preserves agreement, but agreement of a projection does
 246 not establish agreement of the full vector.

247 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 248 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 249 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 250 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

251 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

252 duce an initial mismatch. Approximate observations leave a nondecaying
 253 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 254 update to the phase process.

255 4.3 What approximate agreement determines

256 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 257 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 258 Weights that vanish outside their patches and sum to one at each site select
 259 a global profile by weighted averaging. The selection lies within ε of each
 260 patch there, and any two such selections differ by at most ε , independently
 261 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 262 $L\varepsilon$ across selections.

263 At zero tolerance, the selection is the unique exact extension. At every
 264 positive tolerance there are families with no exact extension; selection then
 265 gives an approximation whose dependence on weights is bounded. Supple-
 266 mental Material gives a three-site witness attaining the bound. The mass-
 267 profile construction uses the scalar case of the content dynamics and addi-
 268 tionally requires nonnegativity of the initial profiles and observations. The
 269 general vector result supplies no identification with the spatial-measure sheaf.

270 These constructions concern declared objects: a specified cover, contents
 271 valued in a normed space, and encoders related on overlaps. They select
 272 neither the neural cover nor the decoder. Identifying their contents with
 273 perceptual content, and establishing that the identification bears on experi-

274 ence, is a further modelling task; Section 7 sets out what a measurement of
 275 it would have to establish.

276 **5 Self-representation: a state that models it-** 277 **self**

278 **5.1 The reflexive construction**

279 Compatibility gives a global state. Reflexivity relates it to an encoding within
 280 that state. Equip global sections with a complete metric. If the additional
 281 restriction-resonance condition identifies encoding with restriction to a region
 282 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

283 The region and readout are fixed across the state space. If F contracts and
 284 fixes the section s_* formed from the local descriptions, Banach’s theorem
 285 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 286 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 287 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 288 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 289 obligation.

290 In the cup example, the global state includes the represented scene and the

291 perceiver's relation to it. A local encoding must retain enough information
 292 for the readout to recover that modelled state; cup position alone would be
 293 insufficient if hand configuration and viewpoint could not be reconstructed.

294 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 295 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

296 At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
 297 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 298 weakens.

299 5.2 Reconstruction across a family of states

300 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 301 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 302 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

303 Thus M relevant states pairwise more than 2ε apart require at least M
 304 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 305 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 306 existing self-prediction map.

307 This is an additional accuracy criterion on a family, with no requirement
 308 for multiple exact fixed points. A blind encoder or constant readout fails it

309 for separated states despite having a unique fixed point.

310 Where the encoding is read as the state’s own restriction to a region, the
311 criterion becomes a condition on the region. Restriction resonance requires
312 the encoder and that restriction to identify the same pairs of states, so a
313 boundary reconstructing two relevant states more than 2ε apart while its re-
314 gion reads them identically is not resonant (`not_isRestrictionResonance_`
315 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
316 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
317 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
318 cally is fixed by where the region sits, before an encoding is chosen, so the
319 resolution the family demands is the region’s and no readout supplies it. The
320 implication runs one way: a region that does resolve the family is not thereby
321 shown to carry a resonant encoding, and E89 supplies no such resolution.

322 **5.3 Agreement after a change takes communication**

323 Snapshot compatibility does not specify how quickly descriptions respond
324 when a shared quantity changes. In a synchronous network whose updates
325 read only the site’s own state and declared neighbours’ messages, two initial
326 configurations agreeing on a site’s causal past through T rounds produce the
327 same state there at round T . If two fresh interventions outside that past
328 require reports separated by more than 2ε , no report from that site can be
329 within ε of both. This follows from the reconstruction bound in Section 5.2.

330 A three-site delay line witnesses failure after one round and exact report-

ing after two; cutting the path prevents delivery at every round. A report
 may still happen to be correct about one fixed world. The bound concerns
 responsiveness across interventions.

Throughout this section *prediction* names a reconstruction map; reading
 it as temporal neural prediction would require dynamics. The relevant family
 of states, the metric, the tolerance, the communication graph and the dead-
 line are all declared inputs, as are the represented variables and the fidelity
 counted as accurate. Reading the resulting code count as physical memory
 capacity requires a hardware model, and a readout that varies with the state
 requires a further one.

6 Thermodynamics and composition

The phase and representation results concern coupling, content disagreement
 and reconstruction error. Thermodynamics constrains the physical opera-
 tions that could implement them, and it does so from the side rather than
 along the spine: no edge below derives coherence, compatibility or reconstruc-
 tion from a heat bound, and the resource condition these edges terminate in
 is necessary without being restrictive for any macroscopic candidate (Sec-
 tion 3.2). What this half contributes is a ledger in which a resource claim
 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field

352 reaching disconnected vacuum components (Kibble, 1976). The composition
 353 connects these objects through the explicit hypotheses in Table 1.

354 **6.1 Feedback changes the information constraint**

355 For passive memory X_t driven by an environmental signal S_t , the predictive
 356 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

357 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 358 et al., 2012). Zero retained information also satisfies it, so a task and a
 359 learning rule require additional specification. Relating this bound to Lan-
 360 dauer heat requires a common thermal scale and budget, and memory waste
 361 bounded by erased entropy.

362 An acting agent changes the signal it will receive. Specify policy, world,
 363 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \quad (12)$$

364 The second conditional term vanishes in the passive model. With feedback,
 365 the signed difference can be negative as action creates correlations with the
 366 future. A causal reading requires an environmental state sufficient to describe
 367 the relevant dynamics.

368 For a controlled substep, the active thermodynamic bound instead uses

369 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F||P_R) =$
 370 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 371 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 372 2013). Every forward path of positive probability must have a positive-
 373 probability reverse for this finite identity. A missing reverse gives infinite
 374 divergence. The supplement develops the support conditions and composes
 375 actuation with memory updates on their shared intermediate law.

376 **6.2 Learning and repeated operation need explicit re-** 377 **sources**

378 The finite examples distinguish selecting a feasible policy, relaxing a register
 379 toward an encoded reward, and learning an unknown task from observa-
 380 tions. The observational learner acquires correlation with an environmental
 381 parameter while its action marginal stays uniform. A continuing model exe-
 382 cutes actuation, observation-driven learning and reset in sequence, preserving
 383 above-chance performance at a positive work cost per cycle.

384 A phase-derived sensor makes the informational distinction concrete: an
 385 informative readout and a blind control have the same phase order but dif-
 386 ferent learning performance. Attaching the learner’s register to the actuator
 387 of Section 3.2 ties reward, installed kernel and work to its own executed law.
 388 The learning witness rises above the necessary installed-energy bound while
 389 its matched control remains on the exclusion line. These finite constructions,

390 detailed in Supplemental Material, leave the evolving feedback between phase
 391 dynamics, installation and sensing to be specified.

392 For that model, common energy E , thermal scale θ , local detailed balance
 393 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

394 where expected cumulative work stays within the supplied allowance B at
 395 every prefix. This bounds expectations. Separate constructions track a store
 396 on each trajectory, include finite sources, charge sensor clearing and prepend
 397 preparation. They specify when a mean allowance represents an available
 398 resource; an expected budget alone permits overdrawn paths.

399 These constructions establish costs for their declared processes. A finite
 400 source with a positive net cost permits only finitely many funded cycles,
 401 while continuing operation requires replenishment. Register–bath witnesses
 402 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply
 403 automatically.
 404
 405

406 **6.3 What the composition establishes**

407 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named
 408

409 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

410 with B the register’s heat. The upper allocation follows from a ledger whose
 411 operations deliver that total heat to one reservoir and whose other operations
 412 do not increase joint entropy. Those ledger assumptions are physical inputs.
 413 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.

Table 1 (continued)

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are inde-

422 pendent roots rather than a step from one to the other; both E45 discharges
 423 take their convergence from refinement data and not from the thermody-
 424 namic bound; E78 states that coherence relaxes the supplied cover, and the
 425 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
 426 coders already shared on overlaps. The inferential content of the composition
 427 is its last step, where contraction and a fixed glued section give uniqueness.
 428 What composing them establishes is that the premises are consistent, named
 429 in one place and machine-checked together, and not that coherence follows
 430 from installed energy or self-representation from coherence.

431 Joint witnesses combine a register, boundary, refining mesh and small
 432 spatial cover; active witnesses add controlled thermal updates and a priced
 433 installation. They establish satisfiability under their stated identifications.
 434 Mesh convergence follows from refinement data, and the witnesses establish
 435 no common cortical mechanism. Supplemental Material maps the claims to
 436 their Lean declarations and records the counterexamples. The development
 437 declares no additional axioms beyond its libraries' foundational axioms.

438 **7 What the observations establish**

439 The composition holds under its premises. Whether observation can estab-
 440 lish those premises is a separate question, and the three requirements it
 441 connects concern different objects: phase dynamics, local descriptions and
 442 internal encodings. Their observables must retain these distinctions. The

443 following controls show which mechanisms remain indistinguishable, which
 444 temporal approximations fail, which states one observable cannot separate,
 445 and what independent content measurements must establish. Each is a finite
 446 numerical regime on a declared model, with full parameters and controls in
 447 Supplemental Material, and none of them calibrates a cortical quantity; the
 448 scope notes below concern what each result does and does not separate.

449 **7.1 Stationary distributions leave the mechanism un-** 450 **determined**

451 Equation (2) gives a concentration–coherence relation with no fitted shape
 452 parameter. Estimate r from circular resultant length and a from density
 453 shape: inverting R from the measured r would make agreement automatic.
 454 Even independently estimated agreement identifies a statistical family shared
 455 by different mechanisms.

456 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 457 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 458 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 459 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 460 and diffusion D_0 have that same density. Their stationary distributions co-
 461 incide window by window despite different causes.

462 Resolved dynamics supply additional information. In the ideal model with
 463 continuous paths, quadratic variation accumulates at rate $2D$ and conditional

464 increment means constrain drift. Separating endogenous coupling from a
 465 matched common drive additionally requires a measured drive or controlled
 466 perturbation; identifying electromagnetic coupling requires evidence specific
 467 to that interaction. Supplemental Material gives the equivalence construction
 468 and its numerical check.

469 **7.2 Finite ramps change the observable onset**

470 The square-root law in Eq. (3) describes stationary solutions. A changing
 471 coupling requires time for the phase distribution to approach those solutions.
 472 In a deterministic mean-field control that crosses the threshold symmetrically,
 473 none of the scanned rates meets the declared tracking tolerance: slowing
 474 the crossing does not remove the terminal residual under that initialization.
 475 A slow control variable therefore supplies no general guarantee of branch
 476 tracking at onset. Supplemental Material gives the residual, rate range and
 477 admissible rates away from threshold.

478 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 479 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 480 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 481 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 482 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 483 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 484 onset exponents differ from those of the exact stationary branch evaluated
 485 on the same grids. At the fastest speed the ensemble mean never reaches the

486 fit window, so its fitted slope does not estimate stationary onset.

487 Delayed passage through a noisy bifurcation (Berghlund & Gentz, 2002)
488 makes the apparent exponent depend on ramp speed and fitting window.
489 The specified deviation-based critical ramp speed is not bracketed by this
490 sweep. With fluctuating coupling, the relevant stationary comparison also
491 depends on rate: the mean coupling approximates order in the tested fast
492 regime, while averaging stationary order over coupling works for the tested
493 slow drives confined to one side of threshold. Slow crossing drives can satisfy
494 neither approximation.

495 **The prediction.** The delay scaling is what this section supports as a pre-
496 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
497 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494].
498 The reference for that exponent is the same estimator applied to a deter-
499 ministic mean-field run seeded at the fluctuation floor, which returns 0.425
500 over the same legs at the same escape criterion, inside the interval on the
501 measurement. The distance from 1/2 is what an absolute escape criterion
502 costs: the run crosses the saturation between the floor it starts from and the
503 level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490.
504 Stated as a law rather than as a number: onset delay scales as the square
505 root of the rate at which coupling crosses threshold. Testing it requires only
506 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
507 — so it is a within-subject design that varies the rate of emergence from

508 anaesthesia alone, with each participant as their own control. Such a design
 509 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
 510 meets a size-dependent critical fluctuation floor, so the criterion has to be
 511 referred to each recording's own pre-threshold fluctuation scale rather than
 512 to a shared number. The diffusion D is the remaining input, and Section 7.1
 513 notes that it is estimable from resolved phase traces through the rate $2D$
 514 at which quadratic variation accumulates, which is a different quantity from
 515 the concentration the propofol exercise could not range over.

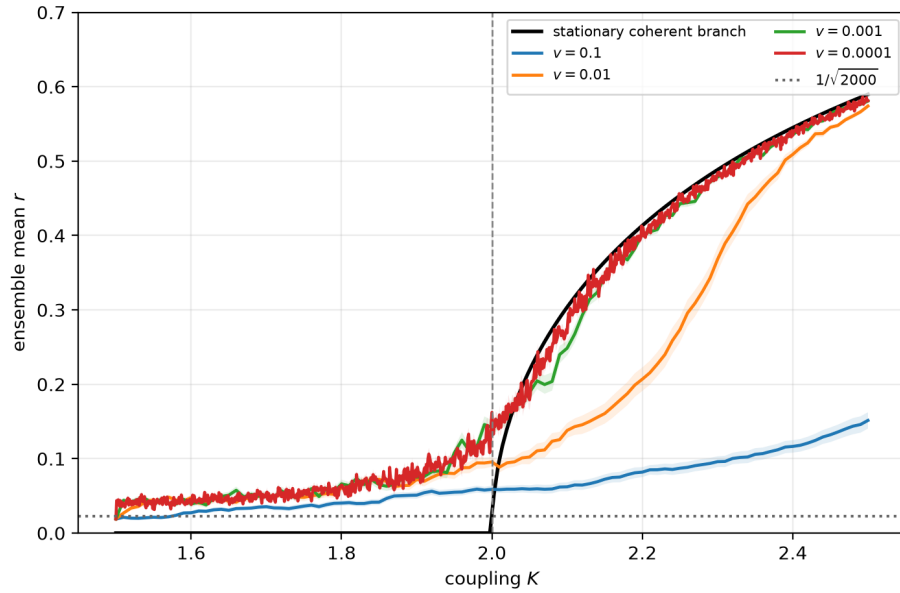


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

516 Additional controls address integration bias, population size and spatial

517 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
518 $[1.958, 2.009]$ contains 2.

519 **7.3 A winding state leaves the order parameter am-** 520 **biguous**

521 The order parameter r is indifferent to where a phase sits in space. A field
522 winding once across a periodic sheet therefore has $r = 0$ while every patch
523 of it stays locked. Such a field is not an artificial construction here. On a
524 torus a winding is stationary for any isotropic symmetric kernel at identical
525 frequencies: the coupling term is odd in the separation against a kernel even
526 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
527 this for any finite periodic array, this sheet included. A winding satisfies the
528 positivity and identical-frequency hypotheses. It violates only the quarter-
529 turn initial spread that the convergence result additionally requires before
530 placing the limit at the uniform state.

531 On the sheet of Section 3's spatial control, a winding survives below
532 an interpolated decay length of 0.1871 mm and unwinds above it. That
533 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the
534 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
535 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
536 order 0.9015 and no winding defect. At fixed K the observable therefore takes
537 two values spanning nearly its whole range. Coupling does not determine the

538 resultant, and a low resultant does not by itself distinguish weak coupling
 539 from a winding field.

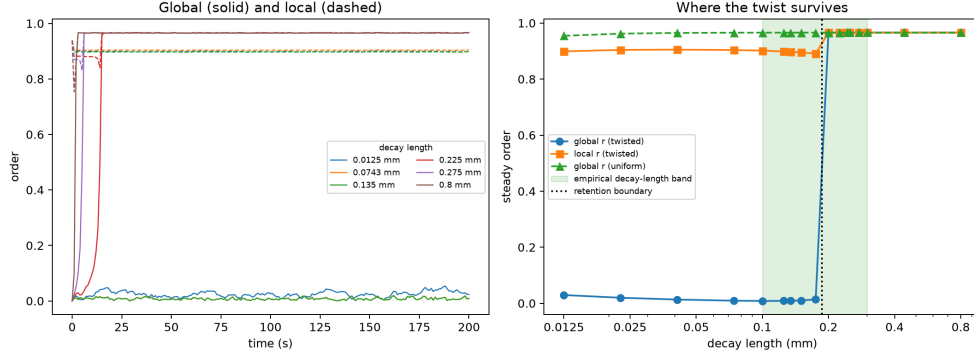


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

540 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 541 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 542 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 543 Heterogeneity shortens the range that sustains a wave without removing it.

544 Which windings persist has a criterion behind it. Linearizing the phase
 545 model about a winding returns a coupling of the same translation-invariant
 546 form, each separation reweighted by the cosine of the phase the winding
 547 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is

548 again even in the separation, so the same characters diagonalize it and ev-
 549 ery mode's rate is one of its eigenvalues. Where the winding advances by
 550 at most a quarter turn across every separation the coupling reaches, the
 551 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 552 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 553 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 554 longer separations is what carries a phase advance past a quarter turn, and
 555 that is the mechanism the decay-length sweep varies. The criterion bounds
 556 rates rather than trajectories: asymptotic stability is not proved, so the
 557 states that persist occupy a narrower range than it admits, and the bound-
 558 ary quoted above is measured and not derived from it.

559 The same reweighted kernel measures the potential's curvature. A wind-
 560 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 561 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 562 tial data does not merely decline to cover a winding, it must exclude it. Along
 563 a mode, the potential's second variation is that kernel's quadratic form, so a
 564 growing mode is a direction of falling potential and a winding carrying one
 565 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

566 An amplitude layer decides which of these states exist, and leaves what
 567 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 568 ery oscillator sits on the unit circle, so a field with an amplitude node has
 569 no representation in these dynamics. The real Stuart–Landau extension
 570 supplies one on the same coupling, and leaves intact the gradient struc-

ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`
`kuramoto`). The phase model is this one with the amplitude ratios set to
one, rather than a separate object.

What the added degree of freedom decides is which windings exist at all.
On a periodic array of any dimension, this sheet included, a winding requires
 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
summed over separations, with ϕ_d the phase the winding advances across d
(`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
lowers it further. A broad enough kernel therefore removes the winding out-
right. The sweep above reaches that same boundary by a different route:
the phase model retains every winding at every coupling, and loses this one
under integration instead. The two layers bound the same regime from op-
posite sides. A phase recording does not separate a state the amplitude layer
excludes from one the dynamics fails to sustain.

These states are windings and not spirals, and on the sheet that is a
theorem rather than a shortage of dimensions. A cortical phase singularity
is a point at which the amplitude vanishes and around which the phase turns
once, read in practice as the circulation around a small plaquette. The states
above have equal modulus at every site, so that circulation is exactly zero
on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can

the layer host a single defect. Over a periodic sheet the circulations of *any* phase field sum to zero, so defects arrive in cancelling pairs (`no_isolated_defect`). The sweep’s near-zero measured defect density agrees with the first of these and says nothing beyond it.

Two limits bound the layer for a reader of the cortical literature. Relaxational dynamics has no propagation speed, which is the quantity a cortical wave measurement leads with. And $|z|$ is a normal-form radius, which a measured envelope is not; this work identifies the two nowhere. Supplemental Material gives the sweep, its uniform control, the amplitude layer and the standing wave it admits inside the band, together with the remaining limits of both.

A vanishing resultant is the absence of global agreement and not of agreement. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch, which falls to zero as the same winding is spread over more sites (`chord_length_of_char_patch`), while the global form of the same bound is vacuous at $r = 0$. Encoders shared on overlaps therefore still bound content disagreement inside a patch of a state that no global observable calls coherent. The winding number itself stays unconnected to content.

This qualifies what a phase trace can establish. An onset criterion stated in r alone registers a change of basin as readily as a change of coupling. The protocol of Section 8.3 therefore needs a spatial observable beside the resultant.

617 **7.4 Coherent adaptation can lose environmental align-** 618 **ment**

619 An adaptive coupling rule might supply representational structure, but its
620 objective must reward the relevant agreement. For symmetric coupling, pair-
621 wise sine terms cancel in the population mean drift. Minimizing squared drift
622 at that fixed mean therefore rewards equalizing drift across nodes. In the
623 frequency-cluster control, this favors coupling between unlike-frequency clus-
624 ters, whereas the diagnostic environmental template rewards coupling within
625 clusters.

626 The formalization proves descent of the squared-drift objective for a spec-
627 ified gradient flow at fixed phases. Its functional differs from thermodynamic
628 entropy production: at a coherent stationary density the probability current
629 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
630 rial gives the operator, its gradient and the conditions of the descent result.

631 The finite control co-evolves noisy phases and symmetric, nonnegative
632 couplings at fixed total resource. Its template is a diagnostic covariance of
633 latent signals, never a training input. Across three seeds, second-half order
634 remains between 0.9167 and 0.9222, while the within-cluster to between-
635 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
636 Random-update, permuted-gradient and frozen-kernel controls are reported
637 in Supplemental Material.

638 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

7.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

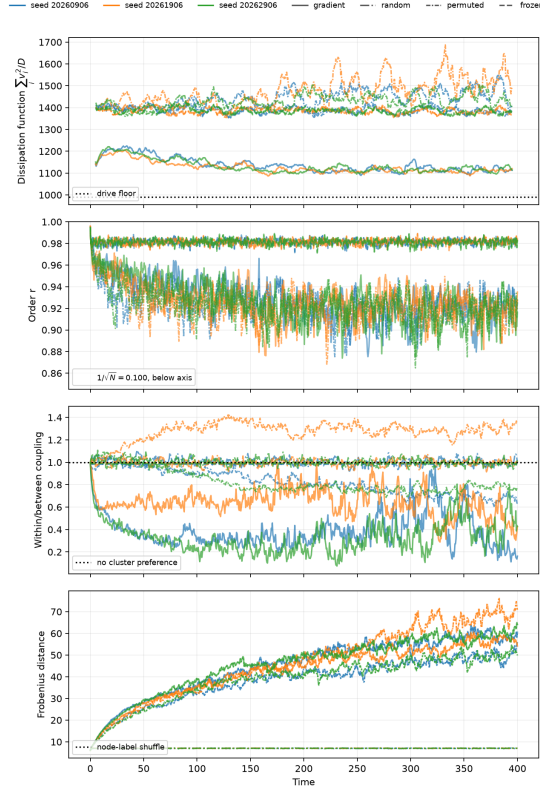


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

661 tent, phase carries a share 1.000 of the content entropy when content is built
662 from it and 0.000 when it is not. A phase-derived statistic reports com-
663 patibility on phase-locked patches whose local descriptions disagree. The
664 implemented content statistic detects their incompatibility. These controls
665 validate a measurement on known constructions; a neural content model and
666 decoder remain to be established.

667 Choosing the cover also chooses a candidate subject: trivial descriptions
668 can agree while carrying little content, and a partition can conceal disagree-
669 ment. The useful choice must therefore be constrained by what neural ac-
670 tivity encodes. The exact composition assumes compatibility of its supplied
671 cover. The content mechanisms above identify additional hypotheses under
672 which agreement can arise; they leave the empirical choice of content and
673 the connection to that cover explicit.

674 **7.6 What an informative phase-density measurement** 675 **requires**

676 The concentration range and the dependence structure decide whether the
677 concentration–coherence relation can be tested at all. Under the specified
678 log-density estimator and bias-floor criterion, discrimination at the observed
679 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
680 designs, rising to 31000 when sites share a mean in groups of 10. These
681 counts describe that estimator and dependence model; the intended recording

682 requires its own calibration. Supplemental Material reports the bin-count
683 controls and the full dependence-aware design.

684 That ceiling comes from exercising the estimators on public propofol-
685 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
686 Bipolar and common-reference montages produce different residuals. The
687 bipolar results are compatible with the Bessel relation over a concentration
688 range too narrow to distinguish it from its linear approximation at the ob-
689 served scatter, which is the range the design scan takes as given. Temporal
690 pooling changes the estimand, and sedation recordings leave the awakening
691 hypothesis untested.

692 An informative dataset therefore needs sufficient concentration range or
693 precision, calibrated estimators and an observation model for dependence,
694 source mixing and reference choice. Public mouse calcium imaging provides
695 thousands of spatially located neurons across waking, sleep and anesthe-
696 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
697 estimation band used here. Site count, state coverage and bandwidth are
698 separate requirements.

699 **8 The cortical hypothesis and an awakening** 700 **test**

701 The cortical proposal identifies a physical source of coupling and asks whether
702 its changes help explain recovery. The tests in Section 7 determine what

703 observations must distinguish before a phase trace can support this identifi-
704 cation.

705 8.1 Physical identification and calibration

706 The cortical hypothesis identifies part of effective coupling with endogenous
707 extracellular electromagnetic interactions. Fields can modulate spike timing
708 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
709 propose coordination of distributed neural information (Pinotsis & Miller,
710 2022; McFadden, 2020). Phase organization under propofol motivates study-
711 ing this regime (Bastos et al., 2021), but does not measure coupling.

712 The formal field predicate requires a jointly continuous kernel, a normal-
713 ized substrate, coupling equal to the kernel average, and no single site pro-
714 viding more than half that strength. Toy constructions satisfy it. A cortical
715 realization must specify the scale at which the kernel and its reduction ap-
716 ply: calcium imaging reports different state-dependent modularity patterns
717 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

718 The spatial reduction is also testable. A row-sum aggregation at zero
719 diagonal tracks the uniform-coupling control over the tested spatial decay
720 range, while frequency heterogeneity displaces the operational threshold. The
721 fit window biases both estimates relative to $2D$; agreement with the control
722 is therefore the supported conclusion. Supplemental Material reports the
723 reduction, its numerical error and the distinct assumptions needed for the
724 kernel-average identification in Eq. (5).

725 The field commitment requires physical evidence beyond a choice of math-
726 ematical representation, and the available support comparisons do not supply
727 it. A finite matrix spread over positive-mass cells preserves its total cou-
728 pling and phase correlation exactly, while point-supported vanishing under
729 an atomless measure and a disadvantage from excluding a better-correlated
730 pair concern specified support constraints; Section 9.3 reads the same com-
731 parisons against a digital candidate. Neither supplies energy prices.

732 A spike-time shift alone supplies no coupling rate. The dimensionless
733 product of population size N , timing sensitivity s , field amplitude E and
734 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

735 This rate describes how induced timing shifts accumulate into drift. Nei-
736 ther it nor the installation factor κ follows from a heat bound. Testing the
737 physical threshold requires independent estimates of the interaction kernel,
738 its aggregation and diffusion in the same cortical regime.

739 8.2 Extracellular geometry and recovery

740 The proposed control variable is extracellular geometry. Extracellular vol-
741 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
742 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
743 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively

744 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 745 an effective-medium description predicts that a larger extracellular fraction
 746 raises conductivity and reduces field amplitude.

747 The coupling-gated hypothesis consequently predicts that extracellular
 748 contraction on awakening raises E and, holding the other factors in Eq. (15)
 749 fixed, raises effective coupling. This argument supplies a direction; the
 750 geometry–coupling relation, its magnitude and its role in human recovery
 751 require measurement.

752 For a smooth crossing at nonzero speed, evaluating the stationary branch
 753 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

754 This follows from Eq. (3) at fixed D and concerns the stationary branch
 755 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 756 dently justified tracking approximation. The crossing controls in Section 7.2
 757 leave that approximation unsupported at onset, even though a separation of
 758 geometry and phase time scales can support it away from threshold. The
 759 awakening hypothesis therefore needs a calibrated dynamical prediction for
 760 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 761 does not supply a recovery latency.

762 **8.3 A protocol for the awakening experiment**

763 The physical test has six requirements:

- 764 1. **Coupling and diffusion in the same units.** Estimate them from re-
765 solved phase dynamics, separating heterogeneity and observation noise.
766 Measure the drive or perturb the system to distinguish endogenous cou-
767 pling.
- 768 2. **Geometry calibrated to coupling.** Validate the geometry observ-
769 able and its relation to effective coupling, with uncertainty. The con-
770 ductivity argument supplies only the expected direction.
- 771 3. **Calibrated phase measurements.** Estimate r by resultant length
772 and a by unweighted regression of log density on centered cosine.
773 Choose site count and binning from a dependence-aware calibration
774 over the intended concentration range. The tested designs require
775 thousands of sites at the observed concentrations; temporal pooling
776 changes the spatial estimand.
- 777 4. **An independent recovery endpoint.** Measure behavioral recovery
778 separately from phase organization.
- 779 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
780 old above the finite-size floor and evaluate its latency under the mea-
781 sured coupling trajectory, including ramp speed or fluctuations. Record

782 a spatial observable beside the resultant, since a winding field reports
783 a low resultant at coupling that sustains local order (Section 7.3).

784 **6. Matched model comparison.** Compare coupling increase, diffusion
785 change, common drive and alternative recovery models under the same
786 observation and filtering assumptions.

787 Failure of the concentration–coherence relation over an informative range
788 constrains the density model. Failure of the calibrated geometry–coupling
789 association constrains the physical gate. Better prediction of independent
790 recovery by a competing mechanism counts against its proposed role. A
791 favorable outcome still leaves the content of coordinated activity to be ex-
792 plained.

793 **9 Discussion**

794 Two things have been established here, and they are of different kinds. The
795 constraints of Section 2 hold outright: synchrony does not deliver compatible
796 content, the resultant does not determine the state, a unique fixed point is
797 not a reconstruction, a region that cannot resolve a family cannot encode it,
798 and agreement about a change takes communication. Any account that reads
799 unity off phase coherence has to meet them, this one included. The account
800 set out in Section 1 is conditional at every step by contrast: each connection
801 in it is a hypothesis supplied rather than a result derived, and Table 1 names
802 all eight. Four of the ingredients assembled there are not standard — the

803 installed-energy condition, which makes a resource requirement necessary for
 804 coherence on the stationary branch; the patch-local form of the coherence-
 805 to-content bound, which is what keeps it from being vacuous; the stability
 806 criterion and amplitude condition that settle when a low resultant reports a
 807 winding field rather than weak coupling; and a machine-checked composition
 808 carrying a witness that discharges all eight hypotheses at once, which is what
 809 makes their conjunction non-empty rather than a list no object satisfies.
 810 What follows takes the conditions in turn, as commitments the proposal
 811 incurs, as measurements that would bear on them, and as questions the
 812 framework leaves open.

813 **9.1 Unity and a minimal self as empirical commit-** 814 **ments**

815 We propose the state glued from compatible local descriptions as a corre-
 816 late of conscious unity, and its accurate internal reconstruction as a correlate
 817 of a minimal self. The motivations are that different contents belong to
 818 one experienced situation and that the situation is experienced from a per-
 819 spective. The conditional results identify relations a realization would need:
 820 sufficient installed resources for the specified phase model, agreement about
 821 shared content, and an encoding that supports reconstruction at the required
 822 fidelity. Discharging E12–E89 for cortex would make the composition a state-
 823 ment about cortex: that a cortical field above threshold relaxes to a cover

824 whose compatible local descriptions glue, and that the glued state is the
825 unique fixed point of a reconstruction map realized there. It would leave the
826 two identifications above exactly where they are, because neither is among
827 the eight and neither follows from their conjunction; they are the commit-
828 ment this subsection names, and Section 9.2 lists the three observations that
829 would count against it.

830 Compatibility allows different descriptions to contribute different infor-
831 mation. It also permits uncertainty and conflicting judgments: apparent cup
832 position and judged cup position can be distinct quantities in a consistent
833 description of their discrepancy. Which quantities are shared must be de-
834 termined independently. Choosing them solely to secure agreement would
835 deprive the proposal of empirical force.

836 The minimal-self proposal concerns the perceiver’s position and condition
837 within the situation, such as the cup’s location relative to a body that can
838 reach it. The encoding could be implicit and need carry no autobiograph-
839 ical narrative. Its proposed necessity is philosophical: even an unreflective
840 episode has a perspective. Gluing by itself requires no self-encoding, and re-
841 constructability alone supplies no functional role. Altering the encoded hand
842 or viewpoint while holding the external scene fixed should change estimates
843 of the perceiver’s relation to the cup as the readout predicts. Whether that
844 represented perspective is experienced remains a further question.

845 **9.2 What a test of the full proposal must establish**

846 Both uniqueness results concern specified objects. They choose neither the
847 cover nor the encoding, and do not select between candidate subjects. In-
848 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
849 here selection requires independent evidence about neural organization. A
850 constant readout can contract and fix a predetermined state while discard-
851 ing its input. The family criterion in Section 5.2 tests the sensitivity that
852 this construction lacks, provided the relevant states and fidelity are declared
853 independently.

854 A test must separately measure coupling relative to diffusion, content
855 compatibility and self-reconstruction, alongside an independent endpoint for
856 experience. Meeting all three without a unified episode would challenge suffi-
857 ciency. A unified episode with incompatible descriptions would challenge the
858 unity identification; one lacking the specified self-encoding would challenge
859 the reflexive requirement. Awakening, dreaming and dissociative states offer
860 possible settings, but absence of report cannot by itself define absence of
861 experience.

862 The next physical requirement is a calibrated geometry–coupling relation
863 and resolved phase dynamics. The installation bound contributes a nec-
864 essary resource condition where its hardware representation applies; cortical
865 mode prices and field-mediated coupling require measurement. For the repre-
866 sentational connection, an informative experiment asks whether a controlled
867 change in coordination alters independently decoded content agreement, with

868 controls for sensory information and decoder quality. The constructed-data
869 estimator establishes feasibility for specified contents. Neural local descrip-
870 tions, shared overlap encoders and the self-encoding have yet to be identified.
871 The population-size dependence of the phase-to-content bound further limits
872 its quantitative use.

873 The separation results explain why these measurements must be inde-
874 pendent. A stationary distribution leaves several recovery mechanisms indis-
875 tinguishable. Equal phase order can accompany incompatible descriptions
876 or different task information. The tested plasticity rule preserves coherence
877 while losing its designated template. Each finding concerns the specified
878 model and observable; their role is to delimit the evidence needed for the
879 proposed connections.

880 **9.3 What the conditions constrain in a digital candi-** 881 **date**

882 The results of Section 2 hold for any candidate, and for a large language
883 model (LLM) running on graphics processing units (GPUs) three of them are
884 checkable rather than merely stateable. That is where this framework bears
885 on claims made for such systems. It does not exclude them, and no result
886 here could; what it does is fix what a claim of unity or self-representation
887 commits its maker to measuring.

888 **Codes, not reports.** Reconstruction across a family is a counting con-
889 straint before it is anything else. If two macrostates the candidate is said
890 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
891 constructing both within ε , whatever the readout. A system’s reports are
892 readouts. Identical verbal reports across two conditions are therefore consis-
893 tent with identical internal codes and settle nothing about the states, while
894 the converse claim — that the system represents its own condition across a
895 declared family — asserts that its internal code separates that family, which
896 is a statement about activations under intervention and not about text. The
897 measurement follows from the statement: declare the family and the fidelity
898 first, then test whether the encoding region distinguishes its members.

899 **The region, not the readout.** Where the encoding is read as the state’s
900 restriction to a region of the machine, the demand falls on the region before
901 any readout exists. For a decoder-only transformer the causal mask makes
902 the network an instance of the update model of Section 5: the activation at
903 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
904 else, so a region’s causal past is fixed by the architecture rather than chosen
905 by interpretation. Two interventions outside that past that demand reports
906 more than 2ε apart cannot both be answered within ε from that region.
907 A claim that a model reports on its own processing consequently carries an
908 obligation to say which positions and layers the report’s causal past contains,
909 and a report that appears to track an intervention outside it indicates a

910 confound or a different region, not a faster channel.

911 **Agreement about what.** Unity requires the candidate’s local descriptions
912 to agree on independently identified shared quantities (Section 4). Common
913 timing, a shared residual stream and fluent output supply none of that, for
914 the reason Section 2 gives: coordination bounds content disagreement only
915 through encoders already related on overlaps, and which quantities two parts
916 of a model describe in common is a decoding question carrying the failure
917 modes of Section 7. The cover is a choice, and choosing it to secure agreement
918 empties the claim.

919 **What is not excluded.** Digital representations can satisfy compatibility,
920 so finite hardware is compatible with the gluing construction and none of
921 the above is a no-go for machines. The coupling-gated route E56–E78 is a
922 different matter: it requires a calibrated physical coupling, its noise regime
923 and a connection from coordination to content agreement, and an attention
924 matrix or a shared clock supplies none of these identifications. Comput-
925 ing a field model likewise leaves its realization in the computing substrate
926 to be established. Those physical obligations fall on the cortical proposal
927 identically.

928 The installed-energy condition of Section 3.2 is restricted in the same
929 way. It constrains a macroscopic candidate only through κ , which is hard-
930 ware data supplied by mode profiles and prices, and Supplemental Material
931 reads the condition against a cortical metabolic budget to return the conver-

932 sion factor it would require rather than a verdict. The hardware comparisons
933 there have similarly restricted scope: fixed wiring can impose a correlation
934 disadvantage, and point-supported kernels vanish under an atomless contin-
935 uum measure, but neither comparison makes the representation conditions
936 unattainable on GPUs. A universal exclusion would need a necessary condi-
937 tion for consciousness and a proof that every relevant implementation violates
938 it, and the present composition supplies neither.

939 Whether experience requires a living substrate is the explicit subject of
940 biological naturalism (Seth, 2026a,b). On that view the properties ground-
941 ing experience are those of self-sustaining organisms, and simulating them
942 does not secure them. The conditions stated here name no biological mech-
943 anism, but computation described as such does not satisfy them either: a
944 candidate must exhibit a physically realized coupling in a measured noise
945 regime, agreement about independently identified shared quantities, and an
946 internal reconstruction meeting a declared fidelity. Whether a realization
947 of those conditions that is not alive would be accompanied by experience
948 is the point at issue, and these results do not settle it. A related position
949 locates the difference in the mode of computation rather than in life as such:
950 biological systems compute across scales that do not separate, and along-
951 side discrete operations carry continuous-valued dynamics belonging to their
952 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
953 here is one measurable instance of that requirement, fixed to a physical field
954 in a declared noise regime, and a description of such a field does not satisfy

955 it.

956 The neuroscientific case against *human-like* consciousness in text-only
957 LLMs concerns embodied sensory interaction, thalamocortical organization
958 and biological regulation linking perception and action to survival (Aru et al.,
959 2023). Their absence supports skepticism, conditional on their causal rele-
960 vance to experience. Those mechanisms are not requirements of the abstract
961 gluing and reconstruction definitions, and workspace-like representations re-
962 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
963 investigate. What this framework adds to that investigation is an order for
964 the questions: declare the family of states and the fidelity, fix the encoding
965 region and read its causal past off the architecture, and only then ask what
966 the system reconstructs. A claim that survives those three steps is a claim
967 about the machine; one that does not survive them was a claim about its
968 output.

969 **9.4 Explanatory scope**

970 Integrated-information, global-workspace and predictive-processing accounts
971 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
972 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
973 candidate physical interaction and conditional relations between local content
974 and self-encoding. Gluing and reconstruction permit multiple substrates;
975 the electromagnetic commitment enters through physical calibration. The
976 formalization checks implications under their hypotheses, and its witnesses

977 establish realizability on declared models. Two things stand outside that
978 result and are not supplied by it: the eight hypotheses are discharged for
979 toy components rather than for cortex, and the identification of the glued
980 state with unity and of its reconstruction with a minimal self is a further
981 commitment that the eight do not contain.

982 The explanatory question is whether the specified relations occur in neural
983 activity and matter for experience. In the cup example, this means establish-
984 ing how coordination helps maintain one represented situation and how the
985 perceiver’s place within it contributes to the episode. The proposal identi-
986 fies candidate correlates and tests of that identification. Why those relations
987 would be accompanied by experience remains unexplained.

988 **Data and code availability**

989 The Lean development, simulation scripts, tests and saved numerical sum-
990 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
991 `cortical_coherence`. Publication simulation values are generated from
992 saved summaries without rerunning production sweeps. The exploratory
993 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
994 processing and selection are described in Supplemental Material. The repos-
995 itory contains the reproducible analysis and does not redistribute the raw
996 recordings.

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1000 **Use of AI tools**

1001 AI assistance to this work was substantial and is not separable by task. Sev-
1002 eral coding assistants and chat interfaces were used in an interleaved way
1003 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1004 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1005 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1006 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1007 accessed with the Hermes CLI. Between them they contributed to the Lean
1008 development, the simulation and analysis code, and the drafting and revision
1009 of both documents. Because the tools were used in combination and in alter-
1010 nation, no section, proof, figure or result is attributable to any one of them,
1011 and none is claimed to be. The author is responsible for all content, including
1012 everything a tool produced. Machine checking establishes the stated Lean
1013 results under their hypotheses; it does not validate biological interpretations
1014 or replace scientific review of generated code and prose.

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